

CSA 1718 – Artificial Intelligence

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Experiment: Prolog Program for Student–Teacher Database

Aim

To write and execute a Prolog program to represent the relationship between **students, teachers, and subjects** using facts and rules.

Objective

- To understand facts and rules in Prolog.
- To represent student–teacher relationships.
- To retrieve information using Prolog queries.

Theory

In Prolog, relationships can be represented using predicates. For example:

- student(Name) represents a student.
- teacher(Name) represents a teacher.
- teaches(Teacher, Subject) represents the subject taught by a teacher.
- studies(Student, Subject) represents the subject studied by a student.

Prolog Program

% Student database

student(ravi).

student(anu).

student(kiran).

student(priya).

% Teacher database

```
teacher(mr_suresh).  
teacher(ms_lakshmi).  
teacher(mr_raj).
```

% Subjects taught by teachers

```
teaches(mr_suresh, ai).  
teaches(ms_lakshmi, dbms).  
teaches(mr_raj, mathematics).
```

% Subjects studied by students

```
studies(ravi, ai).  
studies(anu, dbms).  
studies(kiran, mathematics).  
studies(priya, ai).
```

% Rule to find the teacher of a student

```
student_teacher(Student, Teacher) :-  
    studies(Student, Subject),  
    teaches(Teacher, Subject).
```

Sample Queries

1. Find the teacher of Ravi

?- student_teacher(ravi, Teacher).

Output

Teacher = mr_suresh.

2. Find the teacher of Anu

?- student_teacher(anu, Teacher).

Output

Teacher = ms_lakshmi.

3. Find the students taught by Mr. Suresh

?- student_teacher(Student, mr_suresh).

Output

Student = ravi ;

Student = priya.

4. Find all student-teacher relationships

?- student_teacher(Student, Teacher).

Output

Student = ravi,

Teacher = mr_suresh ;

Student = anu,

Teacher = ms_lakshmi ;

Student = kiran,

Teacher = mr_raj ;

Student = priya,

Teacher = mr_suresh.

Algorithm

1. Start the program.
2. Define student facts.
3. Define teacher facts.

4. Define subjects taught by each teacher.
5. Define subjects studied by each student.
6. Create a rule to connect students and teachers through their common subject.
7. Enter a query.
8. Display the required student-teacher relationship.
9. Stop.

Out Put



The screenshot displays a Prolog IDE with two panes. The left pane, titled 'Program', contains the following code:

```
1 % Student-Teacher-Subject Code Database
2
3 student(lohith, t1, ai101).
4 student(rahul, t2, cs102).
5 student(sai, t1, ai101).
6 student(ankit, t3, db103).
7
8 teacher(t1, revathi).
9 teacher(t2, kumar).
10 teacher(t3, sharma).
11
12 subject(ai101, artificial_intelligence).
13 subject(cs102, computer_science).
14 subject(db103, database_systems).
15
16 % Find teacher of a student
17 student_teacher(Student, Teacher) :-
18     student(Student, TCode, _),
19     teacher(TCode, Teacher).
20
21 % Find subject of a student
22 student_subject(Student, Subject) :-
23     student(Student, _, SCode),
24     subject(SCode, Subject).
```

The right pane shows the execution output:

```
?- sum_to_n(5, Sum).
procedure 'sum_to_n(A)' does not exist

?- knows(ravi, machine_learning).
Sum of natural numbers = 55
procedure 'knows(A,B)' does not exist

?- dob(lohith, DOB).
DOB = date(15,8,2005)
?- student_teacher(lohith, Teacher).
Teacher = revathi

?- student_teacher(lohith, Teacher).
```

Result

The Prolog program was successfully executed to represent and retrieve **student-teacher relationships based on the subjects studied and taught.**